

Complete Module List — Scholarship Management and Verification Portal

This document lists every module in the full system (not just one person's part) — useful for your project report, README, and presentation slide.

Module 1: Authentication & Access Control Module

Purpose: Secure login/registration and role-based access for all three user types.

Features:

- User registration & login (Student, Staff, Admin)
- Password hashing (bcrypt) and JWT-based session tokens
- Role-based route protection (frontend + backend middleware)
- Password reset / forgot password (optional)
- Profile management (view/edit own basic details)

Tech: JWT, bcrypt, Express middleware

Module 2: Student (Applicant) Module

Purpose: Everything a student does — from discovering a scholarship to tracking disbursement.

Sub-modules:

- Scholarship Browse & Search (filter by category, deadline, amount)
 - Application Form (academic details, income, category)
 - Document Upload (marksheets, income certificate, ID proof, bank details)
 - Application Status Tracker (visual stepper across the workflow)
 - Resubmission Handling (re-upload only flagged documents)
 - Application History (past applications across schemes)
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Module 3: Staff (Verification) Module

Purpose: Review, verify, and recommend applications for approval.

Sub-modules:

- Review Queue (applications assigned to staff)
- Document Verification (mark each document valid/invalid/needs resubmission)
- Eligibility Auto-Check (system compares applicant data vs scheme criteria, flags pass/fail with reason)

- Remarks & Comments (per-document and overall)
 - Recommendation Submission (recommend approve/reject → routes to Admin or auto-approves based on scheme config)
 - Staff Performance Stats (applications processed, avg. turnaround)
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Module 4: Admin (Management) Module

Purpose: Configure the program, authorize funds, and oversee everything.

Sub-modules:

- Scholarship Scheme Management (create/edit/close: eligibility rules, amount, deadline, seats, approval-requirement toggle)
 - Final Approval Queue (approve/reject applications recommended by staff)
 - Override Authority (override a staff recommendation with mandatory justification — logged)
 - Disbursement Trigger (mark approved applications as disbursed — mock payment action)
 - Staff Account Management (add/remove/assign staff)
 - Student Account Oversight (view/deactivate accounts if needed)
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Module 5: Workflow & Status Engine Module

Purpose: The automation core — moves applications through states without manual routing.

Features:

- State machine: Submitted → Under Review → Verified → (Pending Admin Approval →) Approved/Rejected → Disbursed
 - Auto-assignment of applications to staff
 - Conditional routing (auto-approve vs admin-approve, based on scheme's `requiresAdminApproval` flag)
 - Automatic status rollback on document rejection (returns to Under Review)
 - Deadline-based auto-closure of schemes
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Module 6: Notification Module

Purpose: Keep every stakeholder informed in real time.

Features:

- In-app notification bell + inbox (read/unread)

- Email notifications (Nodemailer) — submission, verification, approval, rejection, disbursement
 - Deadline reminders (e.g., 3 days before scheme closes) via scheduled job
 - Notification triggers hooked into every status change in the Workflow Engine
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Module 7: Analytics & Reporting Module

Purpose: Turn raw application data into decision-ready insights for Admin.

Features:

- Status breakdown (pie/donut chart)
- Applications-over-time trend (line chart)
- Approval rate by scheme (bar chart)
- Rejection-reason breakdown (bar chart)
- Fund utilization vs budget per scheme (progress bars)
- Staff performance table

Tech: Recharts / Chart.js, aggregation queries (MongoDB aggregation pipeline)

Module 8: Document Management Module

Purpose: Handle secure upload, storage, and retrieval of applicant documents.

Features:

- File upload with type/size validation (PDF/JPG/PNG, max size limit)
- Document preview (inline PDF/image viewer for staff)
- Per-document status tagging (valid/invalid/pending)
- Secure storage (local disk for hackathon scope, or Cloudinary for persistence across deploys)

Tech: Multer, optionally Cloudinary

Module 9: Audit Log Module

Purpose: Full transparency and accountability — who did what, when.

Features:

- Every approve/reject/override/disburse action logged with actor, timestamp, and details
 - Searchable/filterable audit log view (Admin only)
 - Immutable log entries (no edit/delete once written)
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Module 10: Institutional Integration Module (Mocked for Hackathon)

Purpose: Simulate how the portal would connect to real external systems.

Features:

- Mock Student Database lookup (cross-check ID/enrollment number against a dummy institutional record)
- Mock Bank Disbursal API (simulate fund transfer confirmation)
- Designed as a pluggable service layer — real APIs can replace mocks without touching core workflow logic

Module Ownership Map (typical 3-4 person team)

Module	Typical Owner
1. Auth & Access Control	Backend lead / shared foundation, built Day 1 first
2. Student Module	Frontend + backend dev A
3. Staff Module	Frontend + backend dev B
4. Admin Module	Frontend + backend dev C
5. Workflow & Status Engine	Backend lead (shared logic used by Modules 2-4)
6. Notification Module	Whoever finishes core module early, or backend lead
7. Analytics & Reporting	Usually bundled with Admin Module owner
8. Document Management	Bundled with Student Module owner (upload) + Staff Module owner (preview)
9. Audit Log	Bundled with Admin Module owner
10. Institutional Integration (Mock)	Backend lead or whoever has spare time Day 2

How These Map to What You've Already Been Given

- Your personal 2-day plans (Admin/Staff/Student) each cover **Modules 2/3/4 in full**, plus the relevant slices of Modules 6, 7, 8, 9 that belong to that role.
- Modules **1 and 5** are shared foundations — whoever builds Auth and the Workflow Engine should finish them first on Day 1 morning, since every other module depends on them.

- Module **10** is optional polish — only attempt it if your core 9 modules are working end-to-end with time to spare.

This list is what you'd put on a "System Modules" slide/page in your report — it shows judges you decomposed the problem statement completely, not just built a CRUD app.